NEET UG STANDARD — BIOLOGY (PLANT PHYSIOLOGY)

Respiration in Plants

Cellular Oxidation, Glycolysis, Fermentation, Krebs' Cycle, Electron Transport System, Amphibolic Pathway, and Respiratory Quotient

1. Learning Objectives & Chapter Overview

All living organisms require energy to perform vital daily life activities such as absorption, transport, movement, reproduction, and breathing. This energy is obtained by the enzymatic oxidation of macromolecules called food. While green plants and cyanobacteria trap light energy during photosynthesis and convert it into chemical energy stored in the bonds of carbohydrates (such as glucose, sucrose, and starch), non-green parts of plants—as well as heterotrophic animals and saprophytic fungi—depend on translocated or ingested photosynthetic products. Ultimately, all food respired for life processes originates from photosynthesis.

Core Learning Objectives for NEET UG

- Understand the definition of cellular respiration, respiratory substrates, and why ATP functions as the energy currency of the cell.
- Explain how plants manage gaseous exchange through stomata and lenticels without specialised respiratory organs.
- Trace the complete ten-step enzymatic pathway of **Glycolysis (EMP Pathway)** in the cytoplasm, identifying exact steps of ATP utilisation, $\mathrm{NADH} + \mathrm{H}^{+}$ formation, and substrate-level phosphorylation.
- Compare anaerobic pathways—alcoholic fermentation and lactic acid fermentation—with aerobic respiration.
- Analyse mitochondrial aerobic events: oxidative decarboxylation of pyruvate, the **Tricarboxylic Acid (TCA) / Krebs' Cycle** in the mitochondrial matrix, and the **Electron Transport System (ETS)** on the inner mitochondrial membrane.
- Evaluate the chemiosmotic synthesis of ATP via Complex V ($\mathrm{F_0}$–$\mathrm{F_1}$ ATP synthase) and the theoretical assumptions behind the respiratory balance sheet (net gain of 38 ATP).
- Justify why the respiratory pathway is designated as an **amphibolic pathway** and calculate the **Respiratory Quotient (RQ)** for carbohydrates, fats, and proteins.

2. Core Concepts and Detailed Explanations

2.1 Cellular Respiration and Respiratory Substrates

In eukaryotes, photosynthesis takes place inside chloroplasts, whereas the breakdown of complex molecules to yield energy occurs in the **cytoplasm** and in the **mitochondria**. The breaking of the carbon–carbon ($\mathrm{C-C}$) bonds of complex organic compounds through oxidation within the cells, leading to the release of a considerable amount of energy, is called **cellular respiration**.

The organic compounds that are oxidised during this process are known as **respiratory substrates**:

- **Carbohydrates** are the most favoured and commonly oxidised substrates to release energy.
- **Proteins, fats, and organic acids** can also be used as respiratory substrates in some plants under specific physiological conditions.

During intracellular oxidation, the entire energy contained within a respiratory substrate is never released free into the cell at once or in a single step. Instead, it is released in a series of slow, step-wise reactions controlled by specific enzymes and is trapped as chemical energy in the form of **Adenosine Triphosphate (ATP)**. Because energy released by oxidation cannot be used directly by cellular machinery, it is first used to synthesise ATP,

which is later broken down whenever and wherever energy is required. Therefore, **ATP acts as the energy currency of the cell**. In addition to providing energy, the carbon skeletons produced as intermediates during respiration serve as precursors for the biosynthesis of other essential cellular molecules.

2.2 Do Plants Breathe? Gaseous Exchange in Plants

Plants require O_2 for aerobic respiration and release CO_2. Unlike animals, plants possess no specialised organs for gaseous exchange; instead, they rely on **stomata** (in leaves and young green stems) and **lenticels** (in woody stems and roots) for gaseous diffusion. Plants thrive without specialised respiratory organs for three primary physiological and anatomical reasons:

1. **Autonomous Part-wise Gas Exchange:** Each plant part takes care of its own gas-exchange needs. There is very little transport of gases from one plant part to another.
2. **Low Respiratory Rates:** Plants do not present great demands for gas exchange. Roots, stems, and leaves respire at rates far lower than animals do. Large volumes of gases are exchanged only during photosynthesis, and each leaf is well adapted to meet its own needs during these periods. Furthermore, when cells actively photosynthesise, O_2 availability is not a limiting factor because O_2 is evolved right within those cells.
3. **Short Diffusion Distance and Interconnected Air Spaces:** Even in large, bulky plants, the distance that gases must diffuse is small because nearly every living cell in a plant is located close to the surface. In thick, woody stems, the living cells are organised in thin layers inside and beneath the bark, which contains openings called lenticels, while the cells in the deep interior are dead and provide only mechanical support. Furthermore, the loose packing of parenchyma cells in leaves, stems, and roots provides an interconnected network of air spaces, ensuring that most living cells have at least a part of their surface in direct contact with air.

Complete Oxidation (Combustion) of Glucose:

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + \text{Energy}$$

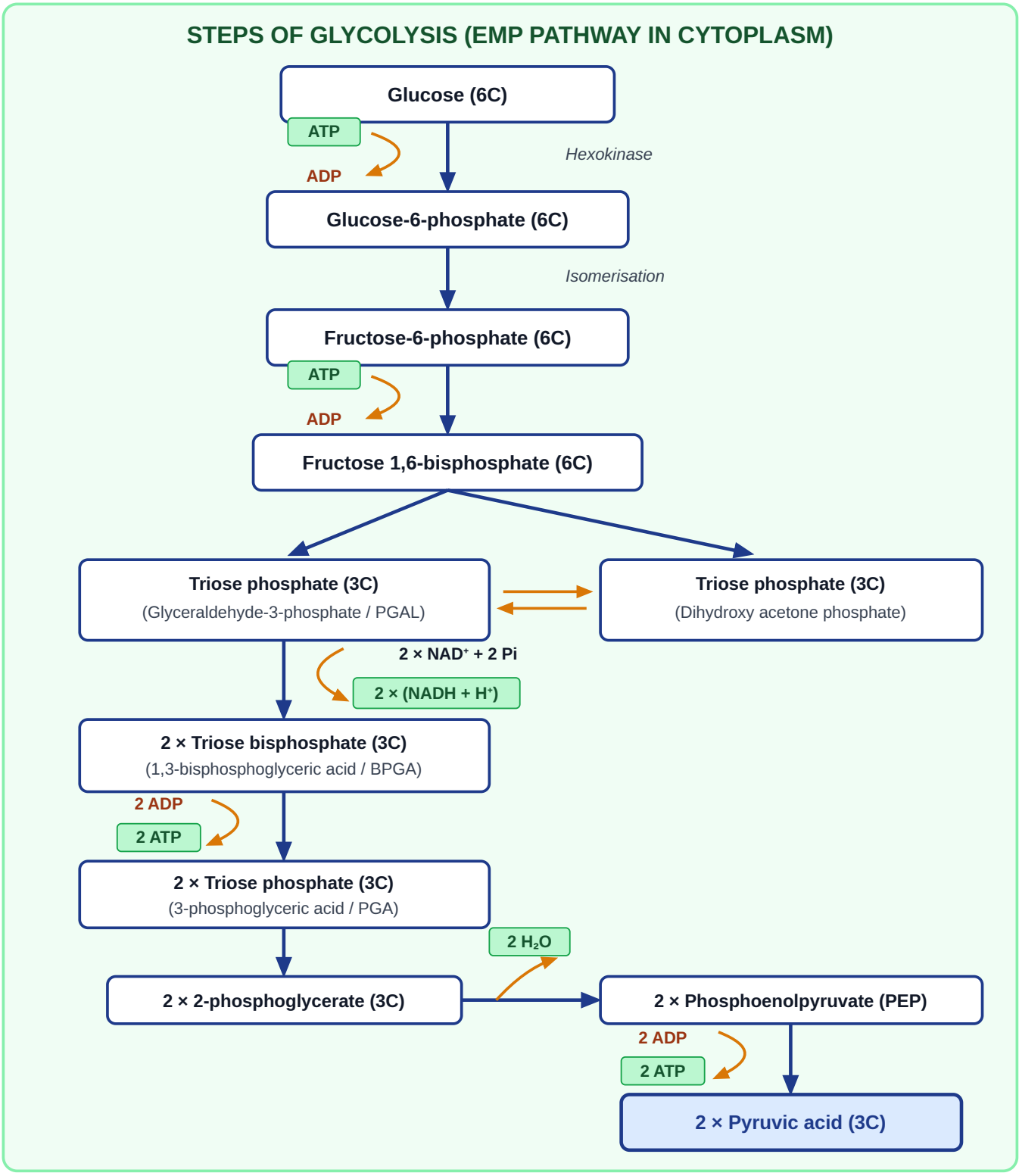
In uncontrolled physical combustion, most of the liberated energy is lost as heat. Plant cells avoid this by catabolising glucose in several small, enzyme-catalysed steps so that individual steps release just enough energy to be coupled to ADP phosphorylation (ATP synthesis).

Although aerobic respiration requires oxygen, the earliest cells on Earth evolved in an oxygen-free atmosphere. Even today, several organisms are adapted to anaerobic conditions—some are **facultative anaerobes** (can respire with or without O_2), whereas others are **obligate anaerobes** (require strictly anaerobic conditions). Crucially, *all living organisms* retain the enzymatic machinery to partially oxidise glucose into pyruvic acid without the help of oxygen. This universal pathway is called **glycolysis**.

2.3 Glycolysis (EMP Pathway)

The term **glycolysis** originates from the Greek words *glycos* (sugar) and *lysis* (splitting). The scheme was elucidated by **Gustav Embden, Otto Meyerhof, and J. Parnas**, and is therefore known as the **EMP pathway**.

- **Site:** Cytoplasm (cytosol) of the cell.
- **Distribution:** Present in all living organisms. In anaerobic organisms, it is the sole process in respiration.
- **Oxygen Requirement:** Does not require oxygen (occurs in both aerobic and anaerobic conditions).
- **Source of Glucose in Plants:** Derived either from **sucrose** (the end product of photosynthesis) or from storage carbohydrates (like starch). The enzyme **invertase** hydrolyses sucrose into glucose and fructose, both of which readily enter the glycolytic pathway.



Tutorial Diagram 1: The Ten Enzymatic Reactions of Glycolysis (EMP Pathway)

Step-by-Step Analysis of Glycolysis

In glycolysis, a chain of **ten reactions** under the control of different enzymes converts one molecule of hexose sugar (glucose, 6C) into two molecules of **pyruvic acid (pyruvate, 3C)**:

1. **Phosphorylation of Glucose (ATP Utilisation #1):** Glucose (and fructose) are phosphorylated to form glucose-6-phosphate by the enzyme **hexokinase** using one molecule of ATP.
2. **Isomerisation:** Glucose-6-phosphate isomerises to produce fructose-6-phosphate. (Subsequent steps of metabolism of glucose and fructose are identical.)

3. **Second Phosphorylation (ATP Utilisation #2):** Fructose-6-phosphate is phosphorylated into fructose 1,6-bisphosphate using a second molecule of ATP.
4. **Cleavage (Splitting):** Fructose 1,6-bisphosphate (6C) is split into two 3-carbon triose phosphate isomers: **3-phosphoglyceraldehyde (PGAL / glyceraldehyde-3-phosphate)** and **dihydroxyacetone phosphate (DHAP)**.
5. **Isomerisation of Triose Phosphates:** DHAP is converted into PGAL so that two molecules of PGAL proceed through the remainder of the pathway.
6. **Oxidation and Phosphorylation ($\mathbf{NADH + H^+}$ Formation):** Each PGAL is converted into **1,3-bisphosphoglycerate (BPGA / 1,3-bisphosphoglyceric acid)**. Two redox-equivalents (in the form of two hydrogen atoms) are removed from PGAL and transferred to a molecule of $\mathbf{NAD^+}$, reducing it to $\mathbf{NADH + H^+}$. Simultaneously, PGAL is oxidised and combines with inorganic phosphate ($\mathbf{Pi}$) to form BPGA. (For 2 molecules of PGAL, $\mathbf{2\,NADH + H^+}$ are formed.)
7. **Substrate-Level Phosphorylation #1 (ATP Synthesis #1):** The conversion of BPGA to **3-phosphoglyceric acid (PGA)** is an energy-yielding reaction; the released energy is trapped by synthesising ATP from ADP (2 ATP formed per glucose).
8. **Intramolecular Rearrangement:** 3-phosphoglyceric acid (PGA) is converted to 2-phosphoglycerate.
9. **Dehydration:** 2-phosphoglycerate loses a molecule of water ($\mathbf{H_2O}$) to form **phosphoenolpyruvate (PEP)**.
10. **Substrate-Level Phosphorylation #2 (ATP Synthesis #2):** PEP is converted into **pyruvic acid**, coupled with the synthesis of a second set of ATP molecules (2 ATP formed per glucose).

> **Energetics of Glycolysis per Glucose Molecule**
>
> - **ATP Utilised (Invested):** $\mathbf{2\ ATP}$ (Step 1: Glucose → Glucose-6-phosphate; Step 3: Fructose-6-phosphate → Fructose 1,6-bisphosphate).
> - **ATP Directly Synthesised:** $\mathbf{4\ ATP}$ ($\mathbf{2\ ATP}$ during BPGA → PGA, and $\mathbf{2\ ATP}$ during PEP → Pyruvic acid).
> - **Net Direct Gain of ATP:** $\mathbf{4\ ATP - 2\ ATP = 2\ ATP}$.
> - **Reducing Equivalents Formed:** $\mathbf{2\,(NADH + H^+)}$ (formed during PGAL → BPGA).

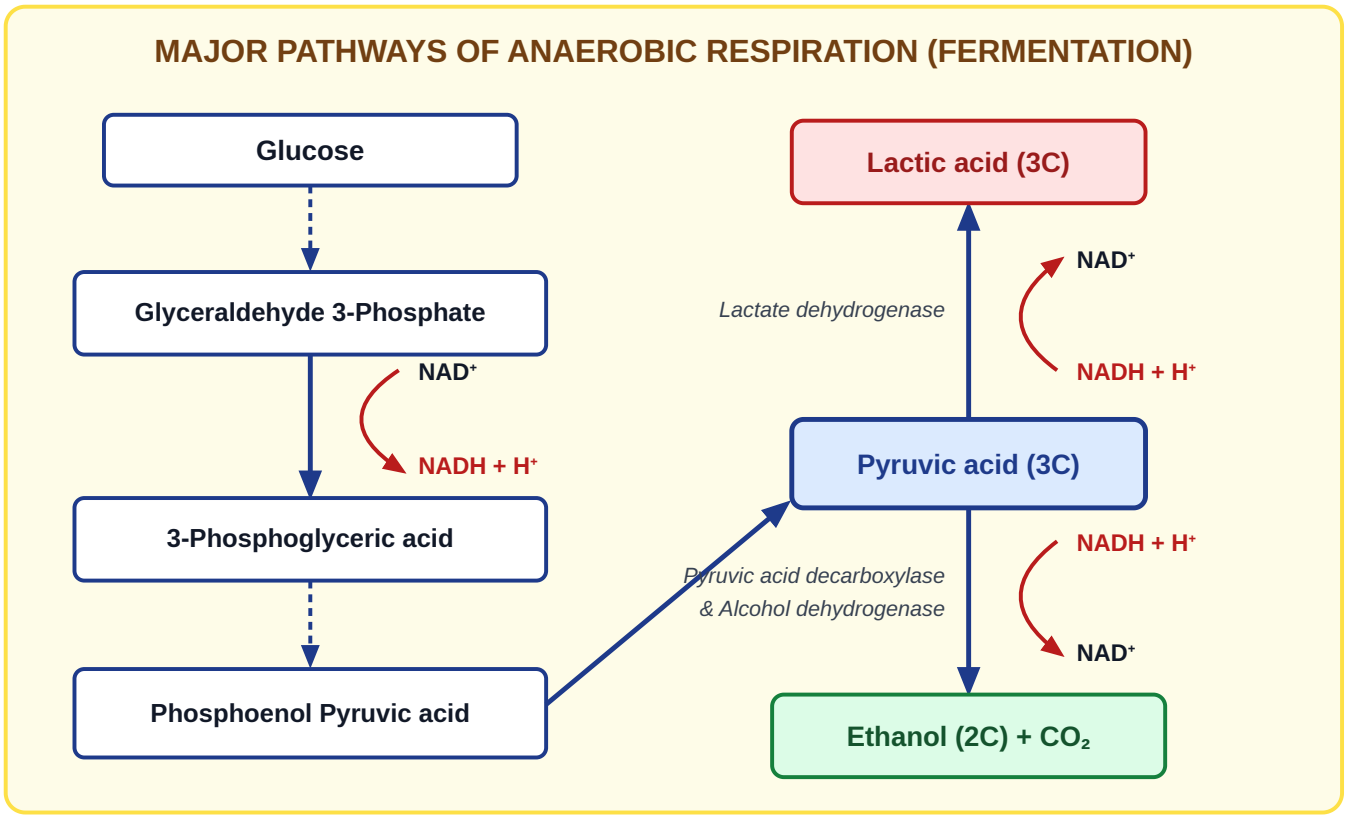
2.4 Metabolic Fate of Pyruvate and Fermentation

Pyruvic acid is the key product of glycolysis. Depending on cellular needs and the availability of $\mathbf{O_2}$, different cells handle pyruvic acid in three major ways:

1. **Lactic acid fermentation** (anaerobic)
2. **Alcoholic fermentation** (anaerobic)
3. **Aerobic respiration (Krebs' cycle)** (requires $\mathbf{O_2}$ supply for complete oxidation of glucose to $\mathbf{CO_2}$ and $\mathbf{H_2O}$)

Fermentation takes place under anaerobic conditions in many prokaryotes, unicellular eukaryotes (such as yeast), and germinating seeds. It involves the incomplete oxidation of glucose:

- **Alcoholic Fermentation (e.g., in Yeast):** Pyruvic acid is converted into $\mathbf{CO_2}$ and **ethanol** in a two-step reaction catalysed sequentially by the enzymes **pyruvic acid decarboxylase** and **alcohol dehydrogenase**.
- **Lactic Acid Fermentation (e.g., in Lactobacillus bacteria and animal skeletal muscles during vigorous exercise):** When oxygen is inadequate for cellular respiration, pyruvic acid is directly reduced to **lactic acid** (without $\mathbf{CO_2}$ release) by the enzyme **lactate dehydrogenase**.



Tutorial Diagram 2: Major Pathways of Anaerobic Respiration (Lactic Acid and Alcoholic Fermentation)

Key Characteristics & Limitations of Fermentation (High-Yield NEET Facts)

- **Role of Reducing Agent:** In both lactic acid and alcoholic fermentation, $\mathrm{NADH + H^+}$ (generated during glycolysis when PGAL is oxidised to BPGA) acts as the reducing agent and is re-oxidised to $\mathrm{NAD^+}$, allowing glycolysis to continue in the absence of oxygen.
- **Low Energy Yield:** Less than **7 per cent** (7%) of the energy in glucose is released during fermentation, and not all of it is trapped as high-energy bonds of ATP.
- **Net ATP Gain:** Only $2\ \mathrm{ATP}$ molecules are synthesised net per glucose molecule fermented to alcohol or lactic acid ($4\ \mathrm{ATP}$ synthesised minus $2\ \mathrm{ATP}$ utilised in glycolysis).
- **Hazardous Products & Yeast Toxicity:** Fermentation is hazardous because either acid or alcohol is produced. **Yeasts poison themselves to death when the concentration of alcohol reaches about 13 per cent** (13%)**.** Consequently, the maximum concentration of alcohol in naturally fermented beverages is about 13%. Alcoholic beverages with an alcohol content greater than 13% are obtained by **distillation**.

2.5 Aerobic Respiration

Aerobic respiration is the process that leads to the **complete oxidation** of organic substances in the presence of oxygen, releasing $\mathrm{CO_2}$, water, and a large amount of energy. In eukaryotes, these post-glycolytic steps take place inside the **mitochondria**. Pyruvate formed in the cytosol during glycolysis is actively transported across the mitochondrial membranes into the **mitochondrial matrix**.

The two crucial events in aerobic respiration are:

1. **Complete oxidation of pyruvate** by the stepwise removal of all hydrogen atoms, leaving **three molecules of** $\mathrm{CO_2}$ (per pyruvate). *Location: Mitochondrial Matrix.*
2. **Passing on of the electrons** removed as part of the hydrogen atoms to molecular $\mathrm{O_2}$ with simultaneous synthesis of ATP. *Location: Inner Membrane of the Mitochondria.*

Link Reaction: Oxidative Decarboxylation of Pyruvate

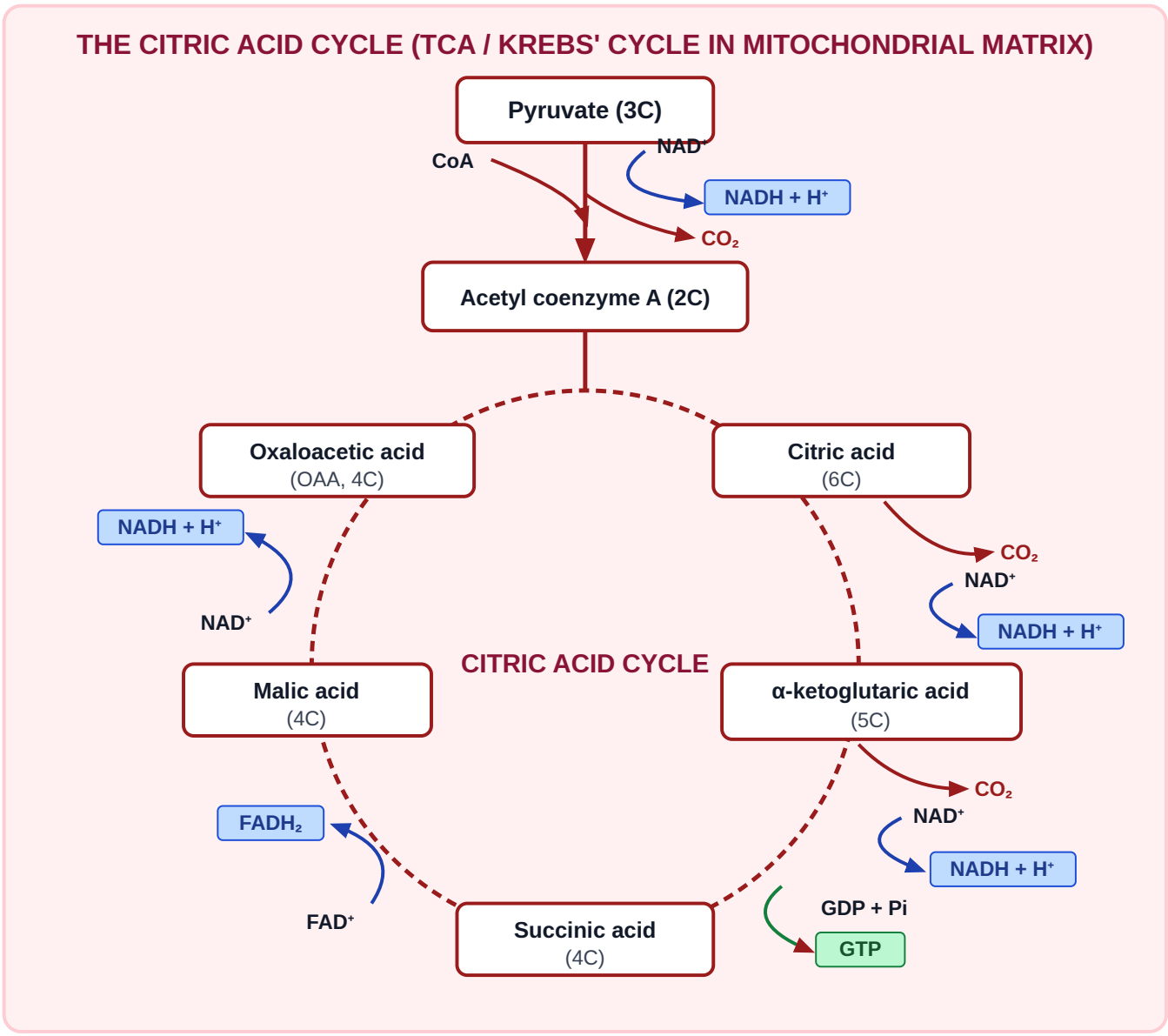
Upon entering the mitochondrial matrix, pyruvate undergoes **oxidative decarboxylation** through a complex set of reactions catalysed by the enzyme complex **pyruvic dehydrogenase (pyruvate dehydrogenase)**. This reaction requires the participation of several cofactors and coenzymes, including $\mathrm{Mg^{2+}}$, $\mathrm{NAD^{+}}$, and **Coenzyme A** (CoA):

$$\mathrm{Pyruvic\ acid + CoA + NAD^{+} \xrightarrow[\text{Pyruvate dehydrogenase}]{\mathrm{Mg^{2+}}} Acetyl\ CoA + CO_2 + NADH + H^{+}}$$

Because one molecule of glucose produces **two molecules of pyruvic acid** during glycolysis, the oxidative decarboxylation of two pyruvates yields $2\ \mathrm{Acetyl\ CoA}$, $2\,\mathrm{CO_2}$, and $2\,(\mathrm{NADH + H^{+}})$.

2.5.1 Tricarboxylic Acid (TCA) Cycle / Krebs' Cycle

Acetyl CoA (2C) enters a cyclic metabolic pathway in the mitochondrial matrix called the **Tricarboxylic Acid (TCA) cycle**, more commonly called **Krebs' cycle** after **Hans Krebs**, who first elucidated it (also called the **Citric Acid Cycle** because citric acid with three carboxyl groups is the first stable product).



Tutorial Diagram 3: The Citric Acid Cycle (Krebs' Cycle) and Link Reaction

Step-by-Step Reactions of the TCA Cycle

1. **Condensation:** The TCA cycle begins with the condensation of the **acetyl group (2C)** (from Acetyl CoA) with **oxaloacetic acid (OAA, 4C)** and **water (H_2O)** to yield **citric acid (6C)**. This reaction is catalysed by the enzyme **citrate synthase**, and a molecule of CoA is released.
2. **Isomerisation:** Citrate (6C) is isomerised to **isocitrate (6C)**.
3. **First Oxidative Decarboxylation:** Isocitrate undergoes oxidative decarboxylation to form **α-ketoglutaric acid (5C)**, releasing the first molecule of CO_2 and reducing NAD^+ to $NADH + H^+$.
4. **Second Oxidative Decarboxylation:** α-ketoglutaric acid (5C) undergoes a second successive step of oxidative decarboxylation to form **succinyl-CoA (4C)**, releasing a second molecule of CO_2 and reducing another NAD^+ to $NADH + H^+$.
5. **Substrate-Level Phosphorylation (Conversion of Succinyl-CoA to Succinic Acid):** Succinyl-CoA (4C) is oxidised/converted to **succinic acid (4C)**, during which a molecule of **GTP (Guanosine Triphosphate)** is synthesised from GDP and Pi. This is a **substrate-level phosphorylation**. In a coupled reaction, GTP is converted to GDP with the simultaneous synthesis of **ATP from ADP**.
6. **Oxidation of Succinate:** Succinic acid (4C) is oxidised to fumarate/malic acid via Complex II (succinate dehydrogenase), where FAD^+ is reduced to $FADH_2$.

7. **Regeneration of OAA: Malic acid (4C)** is oxidised back to **oxaloacetic acid (OAA, 4C)**, reducing a third molecule of $\mathbf{NAD^+}$ to $\mathbf{NADH + H^+}$ and allowing the cycle to continue.

Summary Equation for Mitochondrial Matrix Oxidation (Link Reaction + TCA Cycle per Pyruvic Acid):

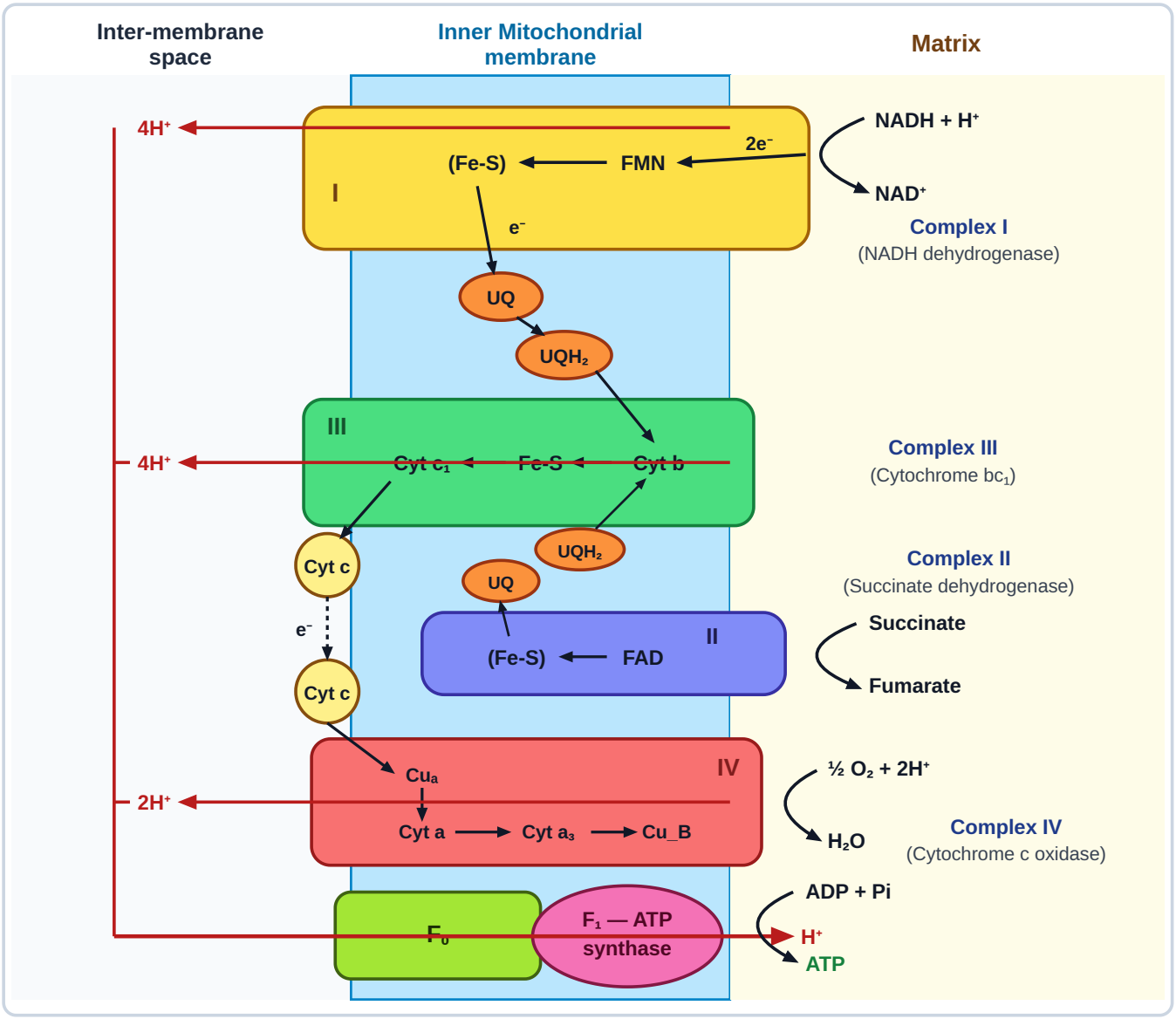
$$\text{Pyruvic acid} + 4\,\text{NAD}^+ + \text{FAD}^+ + 2\,\text{H}_2\text{O} + \text{ADP} + \text{Pi} \xrightarrow{\text{Mitochondrial Matrix}} 3\,\text{CO}_2 + 4\,\text{NADH} + 4\,\text{H}^+ + \text{FADH}_2 + \text{ATP}$$

Key Accounting for TCA Cycle & Link Reaction

- **Per Single Turn of TCA Cycle (per 1 Acetyl CoA):** $3\,(\text{NADH} + \text{H}^+)$, $1\,\text{FADH}_2$, $1\ \text{GTP}\ (\text{or}\ \text{ATP})$, and $2\,\text{CO}_2$.
- **Per 1 Pyruvate (Link Reaction + 1 Turn of TCA):** $4\,(\text{NADH} + \text{H}^+)$ (1 in Link + 3 in TCA), $1\,\text{FADH}_2$, $1\ \text{ATP}\ (\text{via GTP})$, and $3\,\text{CO}_2$.
- **Per 1 Glucose Molecule (2 Pyruvates = 2 Link Reactions + 2 Turns of TCA):** $8\,(\text{NADH} + \text{H}^+)$, $2\,\text{FADH}_2$, $2\ \text{ATP}$, and $6\,\text{CO}_2$.
- **Essential Requirements for Continued TCA Operation:** Continued replenishment of **oxaloacetic acid (OAA)** (the first member of the cycle) and continuous regeneration of NAD^+ and FAD^+ from NADH and FADH_2 respectively via the Electron Transport System.

2.5.2 Electron Transport System (ETS) and Oxidative Phosphorylation

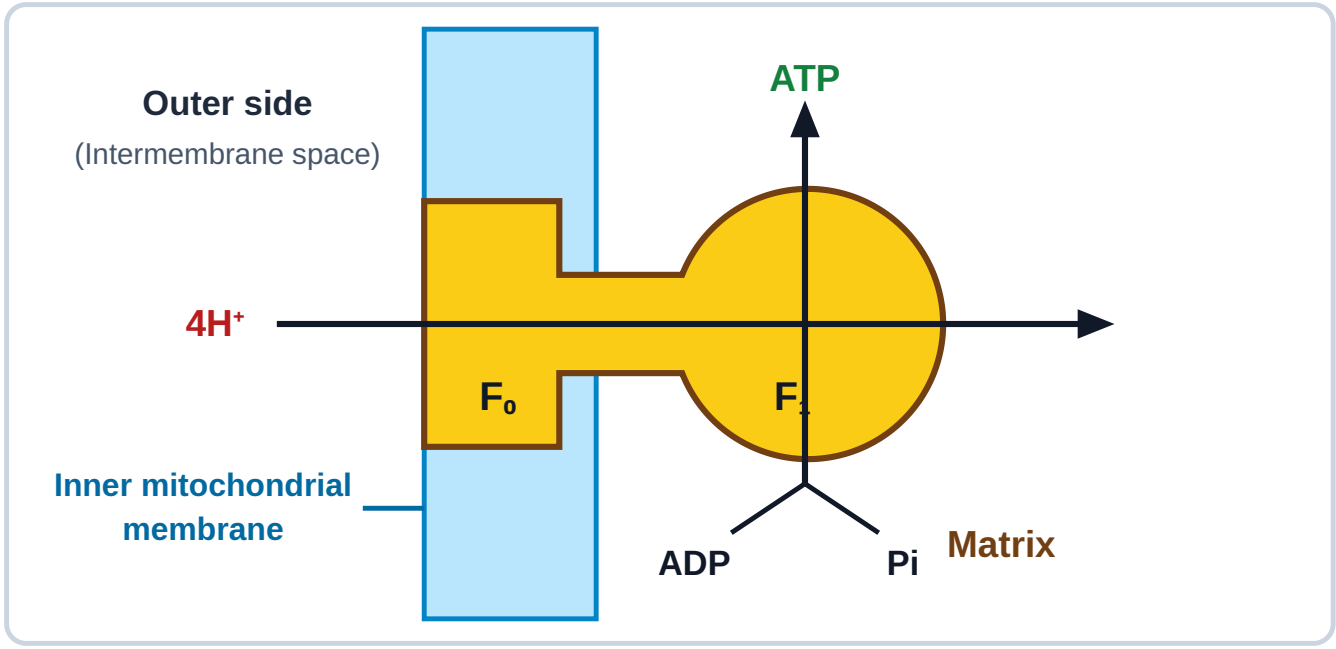
To release and utilise the energy stored in $\text{NADH} + \text{H}^+$ and FADH_2, these reduced coenzymes are oxidised through the **Electron Transport System (ETS)** located in the **inner mitochondrial membrane**, and the electrons are ultimately passed on to O_2, resulting in the formation of H_2O.



Tutorial Diagram 4: Electron Transport System (ETS) and ATP Synthase on the Inner Mitochondrial Membrane

Organisation of the Five ETS Complexes and Mobile Carriers

- **Complex I (NADH dehydrogenase):** Oxidises electrons from $\mathrm{NADH} + \mathrm{H}^+$ (produced in the mitochondrial matrix during the citric acid cycle) via **FMN** and $\mathrm{Fe\text{-}S}$ centres, transferring electrons to **ubiquinone (UQ)** located within the inner membrane.
- **Complex II (Succinate dehydrogenase):** Receives reducing equivalents via $\mathrm{FADH_2}$ generated during the oxidation of **succinate to fumarate** in the citric acid cycle (via FAD and $\mathrm{Fe\text{-}S}$) and transfers them directly to **ubiquinone (UQ)**.
- **Ubiquinone (UQ / Coenzyme Q):** Located within the inner mitochondrial membrane; receives electrons from both Complex I and Complex II, becoming reduced ubiquinone (**ubiquinol,** $\mathrm{UQH_2}$).
- **Complex III (Cytochrome bc_1 complex):** Oxidises reduced ubiquinone (ubiquinol, $\mathrm{UQH_2}$) and transfers electrons (via $\mathrm{Cyt}\ b \rightarrow \mathrm{Fe\text{-}S} \rightarrow \mathrm{Cyt}\ c_1$) to **cytochrome** c.
- **Cytochrome** c**:** A small peripheral protein attached to the **outer surface of the inner mitochondrial membrane**; acts as a **mobile carrier** for the transfer of electrons between **Complex III and Complex IV**.
- **Complex IV (Cytochrome** c **oxidase complex):** Contains **cytochromes** a **and** a_3, and **two copper centres (** $\mathrm{Cu_A}$ **and** $\mathrm{Cu_B}$**)**. It passes electrons to molecular oxygen ($\frac{1}{2}\mathrm{O_2} + 2\mathrm{H}^+ \rightarrow \mathrm{H_2O}$).
- **Complex V (ATP synthase):** Couples the proton gradient generated by electron transport (Complexes I to IV) to the synthesis of ATP from ADP and inorganic phosphate (Pi).



Tutorial Diagram 5: Diagrammatic Representation of ATP Synthesis in Mitochondria via Complex V (F_0–F_1)

Role of Oxygen, Oxidative Phosphorylation, and Chemiosmosis

- **Yield per Electron Donor:** The number of ATP molecules synthesised depends on the nature of the electron donor:
  - Oxidation of 1 **molecule of** $NADH$ gives rise to 3 **molecules of** ATP.
  - Oxidation of 1 **molecule of** $FADH_2$ produces 2 **molecules of** ATP.

- **Vital Role of Oxygen:** Although the aerobic process of respiration takes place only in the presence of oxygen, the direct role of O_2 is limited to the **terminal stage** of the process. Nevertheless, O_2 is vital because it drives the entire process by removing hydrogen from the system; **oxygen acts as the final hydrogen (and electron) acceptor**.
- **Why Called Oxidative Phosphorylation?** Unlike photophosphorylation (where light energy is utilised to produce the proton gradient required for phosphorylation), in respiration it is the **energy of oxidation–reduction** that is utilised to create the proton gradient. Hence, the process is termed **oxidative phosphorylation**.
- **Structure and Mechanism of Complex V (ATP Synthase):** Consists of two major components, F_1 and F_0:
  - F_1 **headpiece:** A **peripheral membrane protein complex** protruding toward the matrix that contains the catalytic site for synthesis of ATP from ADP and inorganic phosphate (Pi).
  - F_0 **component:** An **integral membrane protein complex** embedded in the inner mitochondrial membrane that forms the transmembrane channel through which protons (H^+) cross the inner membrane.
  - **Proton Stoichiometry:** For each ATP produced, $4H^+$ pass through F_0 from the **intermembrane space to the matrix** down the electrochemical proton gradient.

2.6 The Respiratory Balance Sheet

Calculating the theoretical net gain of ATP for every glucose molecule oxidised aerobically yields a net gain of 38 ATP **molecules**. However, this calculation is strictly theoretical and rests on four key assumptions:

1. There is a sequential, orderly pathway functioning, with one substrate forming the next, and with glycolysis, TCA cycle, and ETS pathway following one after another.
2. The $NADH$ synthesised in glycolysis is transferred into the mitochondria and undergoes oxidative phosphorylation.

3. None of the intermediates in the pathway are utilised to synthesise any other compound.
4. Only glucose is being respired—no other alternative substrates are entering the pathway at any of the intermediary stages.

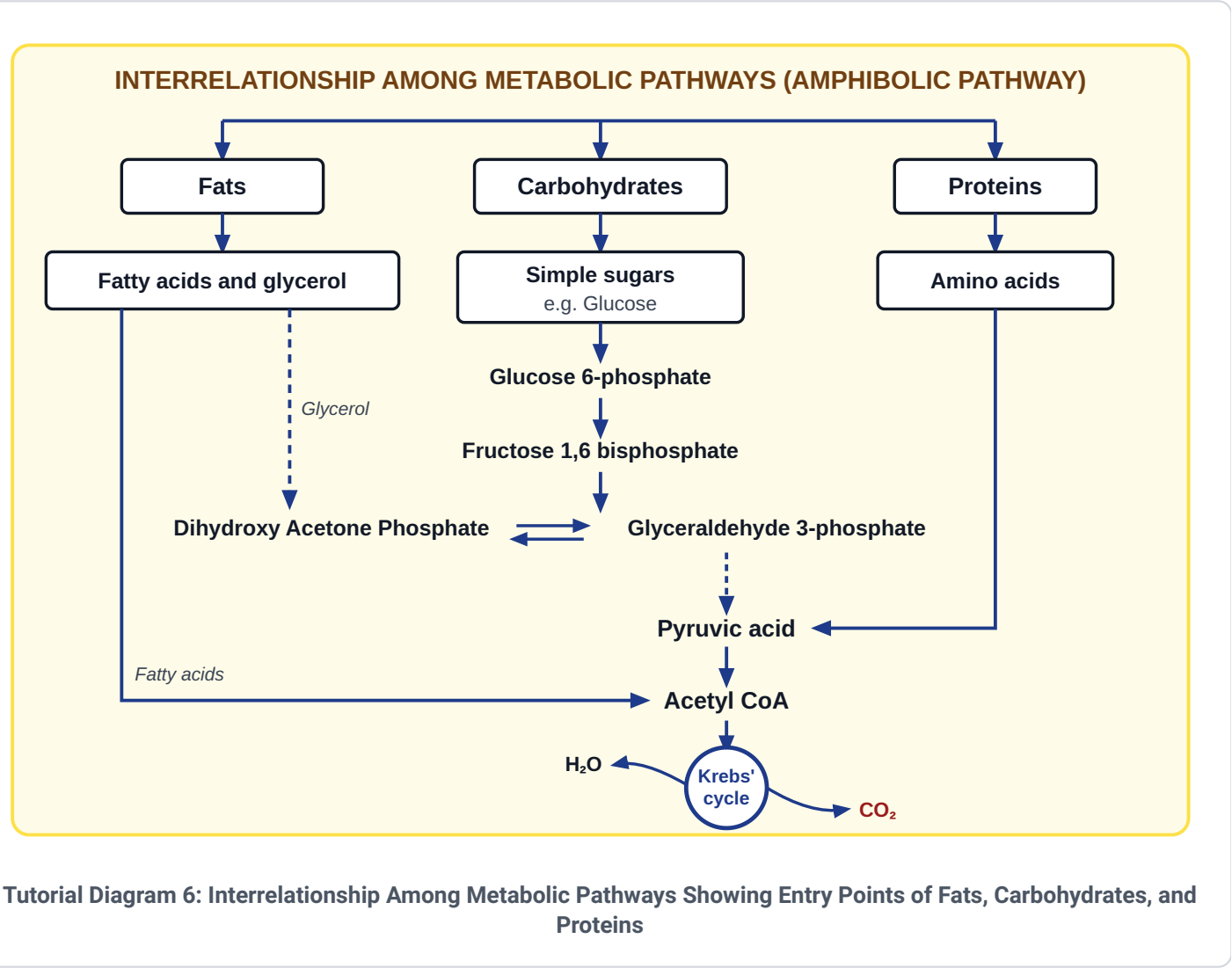
Why these assumptions are not valid in a living system: In reality, all metabolic pathways work simultaneously rather than sequentially one after another; substrates enter and are withdrawn from the pathways as and when necessary; ATP is utilised as and when needed; and enzymatic rates are controlled by multiple regulatory mechanisms.

2.7 Amphibolic Pathway

While glucose is the favoured substrate for respiration (and all other carbohydrates are usually first converted into glucose before entering the pathway), fats and proteins can also be respired, though they do not enter at the first step of glycolysis:

- **Fats:** First broken down into **glycerol** and **fatty acids**.
  - **Fatty acids** are degraded to **acetyl CoA** before entering the pathway.
  - **Glycerol** is converted into **PGAL (glyceraldehyde 3-phosphate / DHAP)** to enter the glycolytic pathway.

- **Proteins:** Degraded by **proteases** into individual **amino acids**. After **deamination**, depending on their carbon skeleton structure, amino acids enter the pathway at some stage within the **Krebs' cycle**, or as **pyruvate**, or as **acetyl CoA**.



Tutorial Diagram 6: Interrelationship Among Metabolic Pathways Showing Entry Points of Fats, Carbohydrates, and Proteins

Why is the Respiratory Pathway Called Amphibolic rather than Purely Catabolic?

Traditionally, respiration was considered strictly a catabolic (breakdown) pathway. However, the exact same respiratory intermediates at which substrates enter for breakdown are also **withdrawn from the respiratory**

pathway for the synthesis (anabolism) of those biomacromolecules. For instance, fatty acids are broken down to acetyl CoA before entering the pathway for oxidation, but when the organism needs to synthesise fatty acids, acetyl CoA is withdrawn from the respiratory pathway. Similarly, respiratory intermediates serve as the link during both the breakdown and synthesis of proteins. Because the respiratory pathway is involved in **both catabolism (breakdown) and anabolism (synthesis)**, it is accurately termed an **amphibolic pathway**.

2.8 Respiratory Quotient (RQ)

During aerobic respiration, O_2 is consumed and CO_2 is released. The ratio of the volume of CO_2 evolved to the volume of O_2 consumed in respiration is called the **respiratory quotient (RQ)** or **respiratory ratio**:

$$\text{RQ} = \frac{\text{Volume of } CO_2 \text{ evolved}}{\text{Volume of } O_2 \text{ consumed}}$$

The value of RQ depends directly on the chemical nature (degree of oxidation/reduction) of the respiratory substrate used:

1. **Carbohydrates ($RQ = 1.0$):** When carbohydrates are used as substrate and are completely oxidised, equal volumes of CO_2 and O_2 are evolved and consumed:

$$C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O + \text{Energy}$$

$$\text{RQ} = \frac{6\,CO_2}{6\,O_2} = 1.0$$

2. **Fats ($RQ < 1$, specifically ≈ 0.7):** Fats contain relatively less oxygen and more hydrogen/carbon than carbohydrates, so they require more O_2 for complete oxidation. For the fatty acid/fat **tripalmitin** ($C_{51}H_{98}O_6$):

$$2\,(C_{51}H_{98}O_6) + 145\,O_2 \longrightarrow 102\,CO_2 + 98\,H_2O + \text{Energy}$$

$$\text{RQ} = \frac{102\,CO_2}{145\,O_2} = 0.7$$

3. **Proteins ($RQ \approx 0.9$):** When proteins are used as respiratory substrates, the RQ is about 0.9 (less than 1).

Important Physiological Note: In living organisms, respiratory substrates are often more than one; **pure proteins or pure fats are never used as respiratory substrates**.

3. Key Equations, Comparative Tables & Important Results

3.1 Comparative Analysis of Respiratory Processes

Feature	Fermentation (Anaerobic)	Aerobic Respiration
Extent of Breakdown	Partial breakdown of glucose (into ethanol + CO_2 or lactic acid).	Complete degradation of glucose into CO_2 and H_2O.
Net ATP Gain per Glucose	Net gain of only 2 ATP molecules (from substrate-level phosphorylation in glycolysis).	Many more ATP molecules generated—up to a net gain of 38 ATP molecules.
Rate of $NADH$ Oxidation to NAD^+	$NADH$ is oxidised to NAD^+ **rather slowly**.	The reaction is **very vigorous** via the mitochondrial ETS.
Cellular Location	Occurs entirely in the cytoplasm.	Starts in the cytoplasm (glycolysis) and completes in the mitochondria (matrix and inner membrane).

Feature	Glycolysis (EMP Pathway)	Krebs' Cycle (TCA / Citric Acid Cycle)
Site in Eukaryotic Cell	Cytoplasm (cytosol).	Mitochondrial matrix.
Pathway Nature	Linear sequence of 10 enzymatic steps.	Cyclic pathway beginning and ending with oxaloacetic acid (OAA, 4C).
Oxygen Requirement	Oxygen-independent (common to both aerobic and anaerobic respiration).	Strictly aerobic (requires O_2-dependent ETS to regenerate NAD^+ and FAD^+).
CO_2 Evolution	No CO_2 is released during glycolysis.	$2\,CO_2$ molecules released per turn ($4\,CO_2$ per glucose in TCA, plus $2\,CO_2$ in link reaction).
Starting & Ending Molecules	Starts with Glucose (6C); ends with 2 Pyruvic acid $(3C)$.	Acetyl CoA (2C) condenses with OAA (4C) to form Citric acid (6C); OAA is regenerated.

Feature	Cellular Respiration	Combustion
Process Control	Biochemical, slow, step-wise oxidation controlled by cellular enzymes at physiological temperature.	Physico-chemical, uncontrolled, single-step burning occurring at high temperatures.
Energy Trapping	Energy is trapped in chemical bonds of ATP for cellular metabolism; only a fraction is lost as heat.	Almost all energy is liberated immediately as heat and light; no ATP is synthesised.

3.2 Complete ATP Balance Sheet per Molecule of Glucose (Aerobic Respiration)

Stage of Respiration	Location	Direct ATP / GTP (Substrate-Level)	Reduced Coenzymes Produced	ATP from ETS (Oxidative Phosphorylation)	Total Net ATP
1. Glycolysis	Cytoplasm	4 produced $-$ 2 used $=$ 2 ATP	$2\,(NADH + H^+)$	$2 \times 3 = 6$ ATP	8 ATP
2. Link Reaction (2 Pyruvates)	Mitochondrial Matrix	0 ATP	$2\,(NADH + H^+)$	$2 \times 3 = 6$ ATP	6 ATP
3. Krebs' (TCA) Cycle (2 Turns)	Mitochondrial Matrix	2 GTP (or 2 ATP)	$6\,(NADH + H^+)$ $2\,FADH_2$	$6 \times 3 = 18$ ATP $2 \times 2 = 4$ ATP	24 ATP
Grand Total per 1 Glucose		4 ATP (Net Direct)	$10\,NADH + H^+$, $2\,FADH_2$	34 ATP (via ETS)	38 ATP

4. Step-by-Step Worked Examples

Worked Example 1: Stoichiometric ATP & Proton Calculation in Mitochondria

Problem: Two molecules of pyruvic acid enter the mitochondrial matrix and undergo complete aerobic oxidation via the link reaction, Krebs' cycle, and the Electron Transport System. Calculate: (a) the total number of $\mathbf{NADH + H^+}$ and $\mathbf{FADH_2}$ molecules generated inside the mitochondria, (b) the number of ATP molecules synthesised specifically through oxidative phosphorylation from these mitochondrial coenzymes, and (c) the total number of protons ($\mathbf{H^+}$) that must pass through the $\mathbf{F_0}$ channel of Complex V to synthesise this oxidative phosphorylation ATP.

Step-by-Step Solution:

1. **Coenzymes per Pyruvate in Mitochondria:** According to the summary equation for mitochondrial matrix oxidation, 1 molecule of pyruvic acid produces $\mathbf{4\,(NADH + H^+)}$ ($\mathbf{1}$ in oxidative decarboxylation + $\mathbf{3}$ in TCA cycle) and $\mathbf{1\,FADH_2}$.
   For $\mathbf{2\ molecules\ of\ pyruvic\ acid}$:

$$\mathbf{Total\ NADH + H^+ = 2 \times 4 = 8\ molecules}$$

$$\mathbf{Total\ FADH_2 = 2 \times 1 = 2\ molecules}$$

2. **ATP via Oxidative Phosphorylation (ETS):** Each $\mathbf{NADH + H^+}$ yields $\mathbf{3\ ATP}$ and each $\mathbf{FADH_2}$ yields $\mathbf{2\ ATP}$:

$$\mathbf{ATP\ from\ ETS = (8 \times 3) + (2 \times 2) = 24 + 4 = 28\ ATP}$$

   (Note: An additional $\mathbf{2\ ATP/GTP}$ are formed by substrate-level phosphorylation during succinyl-CoA $\rightarrow$ succinic acid, bringing total mitochondrial ATP to $\mathbf{30\ ATP}$, of which $\mathbf{28\ ATP}$ come from oxidative phosphorylation.)

3. **Proton ($\mathbf{H^+}$) Requirement at Complex V:** For each ATP produced by ATP synthase, $\mathbf{4H^+}$ pass through $\mathbf{F_0}$ from the intermembrane space to the matrix:

$$\mathbf{Total\ H^+\ required = 28\ ATP \times 4\,H^+/ATP = 112\,H^+}$$

Worked Example 2: Carbon Skeleton Tracking & Decarboxylation Steps

Problem: Trace the carbon count of intermediates starting from a single molecule of sucrose ($12C$) hydrolysed by invertase down to the regeneration of oxaloacetic acid in the TCA cycle. How many total decarboxylation steps occur when one molecule of sucrose is completely oxidised aerobically?

Step-by-Step Solution:

1. **Hydrolysis of Sucrose:** $1\ \text{Sucrose}\ (12C) \xrightarrow{\text{Invertase}} 1\ \text{Glucose}\ (6C) + 1\ \text{Fructose}\ (6C)$. Both monosaccharides are phosphorylated by hexokinase and enter glycolysis.
2. **Glycolysis:** Each hexose ($6C$) splits into two triose phosphates ($3C$: PGAL and DHAP) and yields $2\ \text{Pyruvate}\ (3C)$. Thus, 1 **sucrose** yields $4\ \text{Pyruvate}\ (3C)$ molecules. Zero decarboxylation steps occur in glycolysis.
3. **Mitochondrial Decarboxylations per Pyruvate ($3C$):**
   - *Decarboxylation 1 (Link Reaction):* Pyruvate ($3C$) $\rightarrow$ Acetyl CoA ($2C$) + $1\ CO_2$.
   - Condensation: Acetyl CoA ($2C$) + OAA ($4C$) $\rightarrow$ Citric acid ($6C$) $\rightarrow$ Isocitrate ($6C$).
   - *Decarboxylation 2 (TCA Step):* Isocitrate ($6C$) $\rightarrow$ α-ketoglutaric acid ($5C$) + $1\ CO_2$.
   - *Decarboxylation 3 (TCA Step):* α-ketoglutaric acid ($5C$) $\rightarrow$ Succinyl-CoA ($4C$) + $1\ CO_2$.
4. **Total Decarboxylation Steps for 1 Sucrose:** Each pyruvate undergoes 3 decarboxylation reactions. Since 1 **sucrose** produces 4 **pyruvates**, total decarboxylations $= 4 \times 3 = \mathbf{12\ decarboxylation\ steps}$ (releasing $12\ CO_2$ molecules).

Worked Example 3: Respiratory Quotient (RQ) Analysis

Problem: During an experiment on germinating seeds, two seed lots were studied. Seed Lot A evolved $30\ \text{mL}$ of CO_2 while consuming $30\ \text{mL}$ of O_2. Seed Lot B evolved $51\ \text{mL}$ of CO_2 while consuming $72.5\ \text{mL}$ of O_2. Calculate the RQ of both seed lots and identify the predominant respiratory substrate in each.

Step-by-Step Solution:

- **For Seed Lot A:**

$$\text{RQ}_\text{A} = \frac{\text{Volume of } CO_2 \text{ evolved}}{\text{Volume of } O_2 \text{ consumed}} = \frac{30\ \text{mL}}{30\ \text{mL}} = 1.0$$

  An RQ of 1.0 indicates that **carbohydrates** (e.g., starch/glucose) are being completely oxidised.

- **For Seed Lot B:**

$$\text{RQ}_\text{B} = \frac{51\ \text{mL}}{72.5\ \text{mL}} = \frac{102}{145} = 0.7$$

  An RQ of 0.7 matches the stoichiometric oxidation of **fats / fatty acids (such as tripalmitin)**.

5. Original Practice Questions (NEET UG Standard)

Note: The following questions are original practice items designed strictly to match the format, rigor, and syllabus of the NEET UG examination.

Practice Question 1 — Single-Choice MCQ (Glycolysis)

During the conversion of one molecule of glucose into two molecules of pyruvic acid via the EMP pathway, at which of the following pairs of reactions does substrate-level phosphorylation (direct synthesis of ATP) take place?

(1) Glucose to Glucose-6-phosphate and Fructose-6-phosphate to Fructose 1,6-bisphosphate

(2) 3-phosphoglyceraldehyde (PGAL) to 1,3-bisphosphoglycerate (BPGA) and BPGA to 3-phosphoglyceric acid (PGA)

(3) 1,3-bisphosphoglycerate (BPGA) to 3-phosphoglyceric acid (PGA) and Phosphoenolpyruvate (PEP) to Pyruvic acid

(4) 3-phosphoglyceric acid (PGA) to 2-phosphoglycerate and 2-phosphoglycerate to Phosphoenolpyruvate (PEP)

Correct Answer: (3)

Worked Explanation: In glycolysis, ATP is utilised at two steps (Glucose $\rightarrow$ Glucose-6-phosphate and Fructose-6-phosphate $\rightarrow$ Fructose 1,6-bisphosphate). Direct synthesis of ATP (substrate-level phosphorylation) occurs at two subsequent energy-yielding steps: first, when 1,3-bisphosphoglycerate (BPGA) is converted to 3-phosphoglyceric acid (PGA), and second, when phosphoenolpyruvate (PEP) is converted to pyruvic acid.

Practice Question 2 — Single-Choice MCQ (Gaseous Exchange in Plants)

Which of the following statements is **incorrect** regarding why plants can manage gaseous exchange without specialised respiratory organs?

(1) Each plant part takes care of its own gas-exchange needs, and there is very little transport of gases from one plant part to another.

(2) Roots, stems, and leaves respire at rates far higher than animals do, requiring rapid bulk flow of gases through vascular bundles.

(3) In thick, woody stems, living cells are organised in thin layers inside and beneath the bark, while cells in the interior are dead and provide mechanical support.

(4) Loose packing of parenchyma cells in leaves, stems, and roots provides an interconnected network of air spaces.

Correct Answer: (2)

Worked Explanation: Plants do not present great demands for gas exchange because roots, stems, and leaves respire at rates **far lower** than animals do, and there is very little transport of gases from one plant part to another. Hence statement (2) is incorrect.

Practice Question 3 — Single-Choice MCQ (Fermentation)

When yeast carries out incomplete oxidation of glucose under anaerobic conditions, which of the following accurately describes the enzymes involved, the net ATP gained per glucose molecule, and the alcohol concentration at which yeasts poison themselves to death?

(1) Lactate dehydrogenase; net 2 ATP; about 7 per cent alcohol

(2) Pyruvic acid decarboxylase and alcohol dehydrogenase; net 2 ATP; about 13 per cent alcohol

(3) Pyruvate dehydrogenase and alcohol dehydrogenase; net 4 ATP; about 13 per cent alcohol

(4) Invertase and hexokinase; net 2 ATP; about 20 per cent alcohol

Correct Answer: (2)

Worked Explanation: In alcoholic fermentation by yeast, pyruvic acid is converted to CO_2 and ethanol by the enzymes **pyruvic acid decarboxylase** and **alcohol dehydrogenase**. During glycolysis, $4\ ATP$ are synthesised and $2\ ATP$ are utilised, giving a net gain of $2\ ATP$. Yeasts poison themselves to death when the concentration of alcohol reaches about $13\ per\ cent$ (less than 7% refers to the percentage of energy in glucose released during fermentation).

Practice Question 4 — Single-Choice MCQ (Link Reaction & Krebs' Cycle)

Identify the correct sequence of carbon atoms in the organic intermediates formed during the mitochondrial oxidation of pyruvate:

Pyruvate → Acetyl CoA → Citric acid → α-ketoglutaric acid → Succinic acid → Oxaloacetic acid

(1) $3C \rightarrow 2C \rightarrow 6C \rightarrow 5C \rightarrow 4C \rightarrow 4C$

(2) $3C \rightarrow 3C \rightarrow 6C \rightarrow 5C \rightarrow 4C \rightarrow 4C$

(3) $3C \rightarrow 2C \rightarrow 6C \rightarrow 4C \rightarrow 4C \rightarrow 4C$

(4) $6C \rightarrow 3C \rightarrow 2C \rightarrow 5C \rightarrow 4C \rightarrow 4C$

Correct Answer: (1)

Worked Explanation: Pyruvate is a $3C$ compound. Oxidative decarboxylation removes one CO_2 to yield Acetyl CoA ($2C$). Acetyl CoA condenses with OAA ($4C$) to form Citric acid ($6C$). Successive decarboxylations yield α-ketoglutaric acid ($5C$) and Succinyl-CoA / Succinic acid ($4C$), which is oxidised via Malic acid ($4C$) back to Oxaloacetic acid ($4C$).

Practice Question 5 — Matching Type MCQ (Electron Transport System)

Match the components of the mitochondrial Electron Transport System in **Column I** with their correct description or function in **Column II**:

Column I	Column II
A. Complex I	(i) Cytochrome c oxidase complex containing cytochromes a and a_3, and two copper centres
B. Complex II	(ii) NADH dehydrogenase
C. Complex III	(iii) Succinate dehydrogenase
D. Complex IV	(iv) Cytochrome bc_1 complex

(1) A–(ii), B–(iii), C–(iv), D–(i)

(2) A–(ii), B–(iv), C–(iii), D–(i)

(3) A–(iii), B–(ii), C–(iv), D–(i)

(4) A–(i), B–(iii), C–(iv), D–(ii)

Correct Answer: (1)

Worked Explanation: In the inner mitochondrial membrane ETS: Complex I is NADH dehydrogenase (ii); Complex II is succinate dehydrogenase (iii), which transfers reducing equivalents via $FADH_2$ to ubiquinone; Complex III is the cytochrome bc_1 complex (iv); and Complex IV is the cytochrome c oxidase complex containing cytochromes a, a_3, and two copper centres (i).

Practice Question 6 — Assertion–Reason MCQ (Mobile Electron Carrier)

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Cytochrome c acts as a mobile carrier for the transfer of electrons between Complex III and Complex IV in the mitochondrial electron transport chain.

Reason (R): Cytochrome c is an integral transmembrane protein complex deeply embedded within the lipid bilayer of the outer mitochondrial membrane.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

Correct Answer: (3)

Worked Explanation: Assertion (A) is true: cytochrome c acts as a mobile electron carrier between Complex III (cytochrome bc_1) and Complex IV (cytochrome c oxidase). Reason (R) is false because cytochrome c is a **small protein attached to the outer surface of the inner mitochondrial membrane**, not an integral protein of the outer mitochondrial membrane.

Practice Question 7 — Multiple-Statement Evaluation MCQ (Chemiosmosis & Complex V)

Read the following statements regarding ATP synthase (Complex V) and oxidative phosphorylation in mitochondria:

I. The $\mathbf{F_1}$ headpiece is a peripheral membrane protein complex and contains the catalytic site for the synthesis of ATP from ADP and inorganic phosphate.

II. $\mathbf{F_0}$ is an integral membrane protein complex that forms the channel through which protons cross the inner mitochondrial membrane.

III. For each ATP produced, $\mathbf{4H^+}$ pass through $\mathbf{F_0}$ from the mitochondrial matrix to the intermembrane space down the electrochemical proton gradient.

IV. Oxidation of one molecule of $\mathbf{NADH}$ gives rise to $\mathbf{3\ molecules\ of\ ATP}$, while that of one molecule of $\mathbf{FADH_2}$ produces $\mathbf{2\ molecules\ of\ ATP}$.

Which of the above statements are **correct**?

(1) I, II, and IV only

(2) I, II, and III only

(3) II, III, and IV only

(4) I, II, III, and IV

Correct Answer: (1)

Worked Explanation: Statements I, II, and IV are completely accurate. Statement III is incorrect regarding the direction of proton movement: during ATP synthesis, $\mathbf{4H^+}$ pass through $\mathbf{F_0}$ from the **intermembrane space to the matrix** (not from the matrix to the intermembrane space) down the electrochemical proton gradient.

Practice Question 8 — Single-Choice MCQ (Amphibolic Pathway)

When fats and proteins are used as respiratory substrates in plants, at which specific intermediates do (i) fatty acids and (ii) glycerol enter the respiratory pathway?

(1) (i) Fatty acids enter as Glucose-6-phosphate; (ii) Glycerol enters as Acetyl CoA

(2) (i) Fatty acids enter as Acetyl CoA; (ii) Glycerol enters after being converted to PGAL

(3) (i) Fatty acids enter as Pyruvic acid; (ii) Glycerol enters as Oxaloacetic acid

(4) (i) Fatty acids enter as PGAL; (ii) Glycerol enters as α-ketoglutaric acid

Correct Answer: (2)

Worked Explanation: Fats are first hydrolysed into fatty acids and glycerol. Before entering the respiratory pathway, **fatty acids are degraded to acetyl CoA**, whereas **glycerol is converted to PGAL (3-phosphoglyceraldehyde / triose phosphate)**.

Practice Question 9 — Assertion–Reason MCQ (Respiratory Balance Sheet)

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Calculating a net gain of $38\ \mathrm{ATP}$ molecules for every glucose molecule oxidised aerobically remains only a theoretical exercise and is not strictly valid in a living system.

Reason (R): In a living cell, all pathways work simultaneously rather than sequentially, intermediates are withdrawn for biosynthesis as needed, and alternative substrates can enter the pathway at intermediary stages.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

Correct Answer: (1)

Worked Explanation: The respiratory balance sheet calculation assumes a strictly sequential, closed pathway respiring only glucose with zero withdrawal of intermediates. Because living systems operate pathways simultaneously, withdraw intermediates (amphibolic nature), and regulate enzymes dynamically, the calculation is purely theoretical. Thus both (A) and (R) are true and (R) correctly explains (A).

Practice Question 10 — Single-Choice MCQ (Respiratory Quotient)

Arrange the following respiratory substrates in **increasing order** of their Respiratory Quotient (RQ) values when oxidised aerobically:

A. Tripalmitin (a fatty acid/fat)
B. Proteins
C. Glucose (Carbohydrate)

(1) $\mathrm{C} < \mathrm{B} < \mathrm{A}$

(2) $\mathrm{A} < \mathrm{B} < \mathrm{C}$

(3) $\mathrm{B} < \mathrm{A} < \mathrm{C}$

(4) $\mathrm{A} < \mathrm{C} < \mathrm{B}$

Correct Answer: (2)

Worked Explanation: The RQ for tripalmitin (fat) is 0.7 ($\frac{102\ \mathrm{CO_2}}{145\ \mathrm{O_2}}$), the RQ for proteins is about 0.9, and the RQ for carbohydrates (glucose) is 1.0 ($\frac{6\ \mathrm{CO_2}}{6\ \mathrm{O_2}}$). Therefore, the correct increasing order is $\mathrm{A}\,(0.7) < \mathrm{B}\,(0.9) < \mathrm{C}\,(1.0)$.

6. Common Pitfalls & NEET UG Revision Checklist

Common Exam Pitfalls to Avoid

- **Pitfall 1 — Confusing 7% and 13% in Fermentation:** Less than 7% of the energy in glucose is released in fermentation, whereas 13% is the alcohol concentration at which yeasts poison themselves to death. Never swap these two numbers!
- **Pitfall 2 — Counting CO_2 in Lactic Acid vs. Alcoholic Fermentation:** CO_2 is evolved in alcoholic fermentation (pyruvate $3C \rightarrow$ ethanol $2C + CO_2$), but **no CO_2 is evolved** in lactic acid fermentation (pyruvate $3C \rightarrow$ lactic acid $3C$).
- **Pitfall 3 — Per Glucose vs. Per Pyruvate / Acetyl CoA in Krebs' Cycle:** Always check whether a question asks for yields per *Acetyl CoA* (1 turn: $3\,NADH$, $1\,FADH_2$, $1\ GTP$, $2\,CO_2$), per *Pyruvate* including the link reaction ($4\,NADH$, $1\,FADH_2$, $1\ GTP$, $3\,CO_2$), or per *Glucose* ($2\times$ pyruvate yields).
- **Pitfall 4 — Location of Cytochrome c vs. Ubiquinone:** Ubiquinone (UQ) is a mobile carrier located *within* the inner mitochondrial membrane, whereas Cytochrome c is a mobile protein attached to the *outer surface of the inner membrane* (facing the intermembrane space).
- **Pitfall 5 — F_0 vs. F_1 Orientation:** F_0 is the integral transmembrane proton channel; F_1 is the peripheral headpiece protruding into the mitochondrial matrix where ATP is synthesised.

9-Point Final Revision Recap

1. **Cellular Respiration:** Enzymatic oxidation of $C-C$ bonds of respiratory substrates (mainly glucose) in cytoplasm and mitochondria to trap energy as ATP.
2. **Plant Gas Exchange:** Occurs via diffusion through stomata and lenticels; aided by low respiratory rates, autonomous organ-level gas exchange, surface-proximate living cells, and interconnected parenchymatous air spaces.
3. **Glycolysis (EMP Pathway):** Universal 10-step cytoplasmic pathway converting $1\ Glucose\ (6C) \rightarrow 2\ Pyruvic\ acid\ (3C)$. Consumes $2\ ATP$, directly synthesises $4\ ATP$ (net $2\ ATP$), and generates $2\,(NADH + H^+)$.
4. **Fermentation:** Anaerobic incomplete oxidation in cytoplasm; re-oxidises $NADH + H^+$ to NAD^+ slowly; yields only $2\ ATP$ net per glucose.
5. **Link Reaction:** In mitochondrial matrix;
$$Pyruvate + CoA + NAD^+ \xrightarrow{\text{Pyruvate dehydrogenase, } Mg^{2+}} Acetyl\ CoA + CO_2 + NADH + H^+.$$
6. **Krebs' (TCA) Cycle:** In mitochondrial matrix; Acetyl CoA ($2C$) + OAA ($4C$) + $H_2O \xrightarrow{\text{Citrate synthase}}$ Citric acid ($6C$). Yields $3\,(NADH + H^+)$, $1\,FADH_2$, and $1\ GTP\ (ATP)$ per turn via substrate-level phosphorylation (Succinyl-CoA $\rightarrow$ Succinic acid).
7. **Electron Transport System (ETS):** On inner mitochondrial membrane (Complexes I–IV). O_2 is the final hydrogen/electron acceptor, forming water. $1\,NADH \rightarrow 3\ ATP$; $1\,FADH_2 \rightarrow 2\ ATP$. Complex V ($F_0$–$F_1$) synthesises $1\ ATP$ per $4H^+$ moving from intermembrane space to matrix.
8. **Amphibolic Pathway:** Respiratory pathway serves both catabolism and anabolism; fatty acids enter/exit at Acetyl CoA, glycerol at PGAL, and deaminated amino acids at pyruvate, Acetyl CoA, or Krebs' cycle intermediates.
9. **Respiratory Quotient ($RQ = CO_2\ evolved/O_2\ consumed$):** Carbohydrates $= 1.0$; Fats (tripalmitin) $= 0.7$; Proteins ≈ 0.9. Pure proteins or fats are never used alone as respiratory substrates.